RES-5 susceptibility-ratio: the bare-ratio route FAILS (strong coupling);

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The all-order reduction (res5-allorder-commonmode-bound) gave the closure condition $r_k = a_0\,\chi^{(k+1)}/\chi^{(k)} < \frac{1}{2}$, equivalently $\chi^{(k+1)}/\chi^{(k)} < 1/(2a_0) \simeq 5.2$. The operator directed bounding this ratio. This note evaluates it at the base order and reports an HONEST NEGATIVE on the BARE-susceptibility route, with the reframe that the closure must use the common-mode-subtracted susceptibilities. Scope: the base-order bare ratio and the strong-coupling diagnosis; anchor μ^2, operating endpoint.

2. The bare ratio fails at the base order

The Gaussian-sea density susceptibilities are $\chi^{(2)} \sim 2\int G^2$, $\chi^{(3)} \sim 8\int G^3$ (leading ring), so

$$\frac{\chi^{(3)}}{\chi^{(2)}} \sim 4\,\frac{\int G^3}{\int G^2} = 9.05 \quad > \quad \frac{1}{2a_0} = 5.23, \qquad r_2(\text{bare}) = a_0 \cdot 9.05 = 0.866 > \tfrac{1}{2}.$$

(The combinatorial factor 4 is cumulant-convention-dependent; even at half this value r_2 is marginal, and the qualitative conclusion is robust.) The geometric domination BREAKS at the base order if the BARE susceptibilities are used.

3. The sea is strongly fluctuating (the cause)

The bare ratios grow:

$$\frac{\int G^{n+1}}{\int G^n} \xrightarrow{n} \frac{1}{\hat{r}} \approx 2.5,$$

so the bare cumulants increase $\sim (1/\hat{r})^n$ – the Brazovskii sea is strongly fluctuating (as expected: Brazovskii is a fluctuation-driven first-order transition). A bare-susceptibility geometric bound therefore cannot hold; the bare series is asymptotic / strong-coupling.

4. Reconciliation and the reframe (this does NOT refute RES-5)

This is consistent with SC-SCOPE's $n = 3$ thin endpoint ($\sim 4\%$): SC-SCOPE measured the pattern-dependent DIFFERENCE $\Delta F^{(3)} = F^{(3)}[\mathcal{P}] - F^{(3)}[\mathcal{R}_H]$ (common-mode subtracted), which is $\sim 4\% \ll r_2(\text{bare}) = 0.866$. The strong common-mode part $F^{(n)}_{\text{cm}}(D_0)$ cancels in the difference; only the

a_0-suppressed pattern-dependent residual survives. Therefore the closure condition must be stated on the SUBTRACTED susceptibilities,

$$r_k = a_0 \frac{\chi_{\rm pd}^{(k+1)}}{\chi_{\rm pd}^{(k)}} < \tfrac{1}{2}, \qquad \chi_{\rm pd}^{(k)} = \chi^{(k)} - \chi_{\rm cm}^{(k)},$$

NOT the bare $\chi^{(k)}$. The bare route is eliminated; bounding the SUBTRACTED ratio (after the explicit all-order common-mode subtraction) is the genuine residual.

5. Status and the sharpened residual

T3 / HONEST NEGATIVE on the bare route. Established: the bare ratio $\chi^{(3)}/\chi^{(2)} = 9.05 > 5.23$ fails; the sea is strong-coupling; the closure requires the common-mode-subtracted susceptibilities. **ELIMINATED:** the elementary bare-susceptibility geometric bound. **OPEN (sharpened residual):** the bound on the SUBTRACTED ratio $\chi_{\rm pd}^{(k+1)}/\chi_{\rm pd}^{(k)}$ – a strong-coupling computation requiring the explicit all-order common-mode subtraction of the Brazovskii sea. No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. This is a genuine research frontier, not closeable by elementary susceptibility-ratio bounds.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"The factor* 4 *is convention-dependent – maybe the bare ratio is actually* < 5.2*."* VALID-with-mitigation: at half the factor the bare $r_2 \approx 0.43$ is marginal ($< \frac{1}{2}$ but no margin), and the strong-coupling GROWTH $\int G^{n+1}/\int G^n \to 1/\hat{r} \approx 2.5$ is convention-INDEPENDENT, so even if $n = 3$ were marginal, higher n would breach. The bare route is at best marginal and asymptotically fails.

(β) *"Then the previous reduction note was wrong."* PARTIALLY: the reduction's STRUCTURE (geometric domination of r_k) is correct, but the χ in it must be the SUBTRACTED susceptibility, not the bare one – this note supplies that essential qualification. Not a retraction; a sharpening.

(γ) *"Does this refute RES-5 or B1?"* DISMISSED: it eliminates one ROUTE (the bare-ratio bound) and sharpens the residual to the subtracted-susceptibility bound. B1 stays T6 on {H-LAYER}; no claim is withdrawn (none had been made that the bare ratio < 5.2).

Quantitative sanity check (mandatory): *magnitude* – bare $\chi^{(3)}/\chi^{(2)} = 9.05 > 5.23$ (fails); *growth* – $\int G^{n+1}/\int G^n = 1.32/2.26/2.51 \to 1/\hat{r} = 2.97$ (strong-coupling, convention-free); *consistency* – SC-SCOPE subtracted $n = 3 \sim 4\% \ll$ bare 0.866 (confirms the subtraction is essential); *sign* – the bare cumulants are positive and growing, so the bare geometric tail diverges – only the subtraction saves it.

7. Result footer

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Result ID:              B1-RH-ENUM / RES-5 susceptibility-ratio (bare route
                        eliminated)
Precise statement:      The bare Gaussian-sea ratio chi^(3)/chi^(2) ~ 4 int G^3/
                        int G^2 = 9.05 > 1/(2a0) = 5.23, so r_2(bare) = 0.866 >
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	1/2: the bare-susceptibility geometric-domination route FAILS. The bare ratios int G^(n+1)/int G^n -> 1/r_hat ~ 2.5 (strong-coupling). The closure must use the COMMON-MODE-SUBTRACTED susceptibilities chi_pd^(k) (SC-SCOPE's ~4% n=3 is the subtracted, not bare, value). Bare route eliminated; the subtracted-ratio bound is the residual.
Scope:	Base-order bare ratio + strong-coupling diagnosis; anchor mu^2, endpoint. Convention factor 4 flagged (growth is convention-free).
Dependencies:	res5-allorder-commonmode-bound (the reduction); neargap-common-mode-resolution (the subtraction mechanism); SC-SCOPE (subtracted n=3); Math424-AddA.
Evidence grade:	ANALYTIC + EXECUTED (res5_susceptibility_ratio.py 4/4).
Reproduction command:	python codes/vacuum/res5_susceptibility_ratio.py
Expected output:	bare chi3/chi2 = 9.05 > 5.23; r2_bare = 0.866; growth -> 2.5; 4/4 PASS.
Falsification gate:	A correct cumulant normalisation giving the bare ratio < 5.2 with the growth bounded (would revive the bare route); or a subtracted-susceptibility bound chi_pd^(k+1)/ chi_pd^(k) < 5.2 (would close RES-5).
Tier before / after:	No tier flip. B1 T6 on {H-LAYER}. RES-5: bare-ratio route ELIMINATED; residual sharpened to the subtracted-susceptibility bound (strong-coupling).
No-overclaim statement:	This eliminates the bare-ratio ROUTE (honest negative); it does NOT refute RES-5 or B1, and does NOT close RES-5. The genuine residual -- the common-mode-subtracted susceptibility bound for the strongly-fluctuating Brazovskii sea -- is a research-grade strong-coupling problem. B1 stays T6 on {H-LAYER}.
Next required action:	Construct the all-order common-mode subtraction explicitly (the connected pattern-dependent susceptibilities) and bound chi_pd^(k+1)/chi_pd^(k); this is the genuine RES-5 frontier and warrants a focused effort.